

```

#Om Bhujbal 22CV007
# To find BOD at 7th day 25°C
K1 = float(input("Decay Coefficient at 20°C:"))
T = float(input("Temperature of 3rd day BOD:"))
T1 = float(input("Temperature of 7th day BOD:"))

K2 = (K1 * (1.047) ** (T1 - T))
print("The value of K2 is:", K2)

# Ultimate BOD
e = 2.718

B1 = float(input("BOD at 3rd day 20°C:"))
t = float(input("Time in days for finding B1:"))
E = (1 - e ** (-0.23 * t))
print("The value of E is:", E)

L0 = B1 / E
print("The value of Ultimate BOD (L0) is:", L0)

t1 = float(input("Time in days for finding B2:"))
E1 = (1 - e ** (-K2 * t1))
print("The value of E1 is:", E1)

B2 = L0 * E1
print("The value of BOD at 7th day 25°C is:", B2)

```

```

Decay Coefficient at 20°C:0.23
Temperature of 3rd day BOD:20
Temperature of 7th day BOD:25
The value of K2 is: 0.2893751572825015
BOD at 3rd day 20°C:50
Time in days for finding B1:3
The value of E is: 0.49838804582143437
The value of Ultimate BOD (L0) is: 100.32343355585682
Time in days for finding B2:7
The value of E1 is: 0.8680610647811111
The value of BOD at 7th day 25°C is: 87.08686655499413

```

```

#Om Bhujbal 22CV007
# Determination of density of sludge removed from aeration tank
M = float(input("Enter the value of initial mass:"))
S = float(input("Enter the value of solid content in sludge (%):"))
Gs = float(input("Enter the value of Specific gravity of sludge solid:"))
Rho_W = float(input("Enter the value of density of water:"))

Ws = (S / 100) * M
m = M - Ws
print("The value of mass of water:", m)
print("The value of Solid Content in sludge:", Ws)

Vw = m / Rho_W
print("The value of Volume of water:", Vw)

Rho_S = Gs * Rho_W
print("The value of Density of solid content in sludge:", Rho_S)

```

```
Vs = Ws / Rho_S
print("The value of volume of solid content in sludge:", Vs)

Vt = Vw + Vs
print("The value of total volume of sludge:", Vt)

Rho_SL = M / Vt
print("The value of Density of sludge removed from aeration:", Rho_SL)
```

```
Enter the value of initial mass:100
Enter the value of solid content in sludge (%):2
Enter the value of Specific gravity of sludge solid:2.2
Enter the value of density of water:1000
The value of mass of water: 98.0
The value of Solid Content in sludge: 2.0
The value of Volume of water: 0.098
The value of Density of solid content in sludge: 2200.0
The value of volume of solid content in sludge: 0.0009090909090909091
The value of total volume of sludge: 0.09890909090909092
The value of Density of sludge removed from aeration: 1011.0294117647057
```